



INDUSTRIAL ENGINEERING AND MANAGEMENT

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2025	M.TECH	IIT Kharagpur	8.45 / 10
2023	B.Tech	SHUATS, Prayagraj	8.81 / 10
2017	CLASS XII	Delhi Public School, Bokaro	83%
2015	CLASS X	Shanti Mission Academy, Saharsa	10 / 10

AWARDS AND ACHIEVEMENTS

- Secured **AIR 15** in GATE 2023 for Agricultural Engineering with a score of 707, achieving a top percentile of about **99.358%** nationally.
- Received the Late Rev Gulzari Lal Memorial **Award** in 2022 as recognition for exceptional academic performance during undergraduation.
- Secured **1st Rank** at the 1st Annual Seminar on Green Energy, showcasing strong presentation and comprehension skills.

INTERNSHIPS

Data Science Intern | Ford Business Solutions

[May'24 - Jul'24]

- Built an interactive dashboard using **Python Dash** to evaluate and visualise data quality issues, providing options for handling them.
- Engineered a **Random Forest Regressor** to accurately impute missing values, achieving an **R²** score of **84.09%** and MAE of 139.13.
- Developed a **KNN** model to detect anomalous data due to decimal place error, achieving an **accuracy** of **86.5%** and an **AUC-ROC** of **0.89**.
- Processed and analyzed over **2.5 lakh** records from 2020 to 2023, ensuring robust and reliable model performance in real-world scenarios.

PROJECTS

Forecasting Crude Oil Price in India using SARIMA | Course Project | Dr. B. G. Menon

[Jan'24 - Apr'24]

- Analyzed crude oil price datasets from the Ministry of Petroleum & Natural Gas, identifying seasonality through **decomposition analysis**.
- Performed data preparation using **Shapiro-Wilk** and **ADF tests**, achieving **stationarity** via first differencing and **log transformation**.
- Developed a **SARIMA** model using **Akaike Information Criterion** (AIC), and validated forecasts with **MAE (0.0761)** and **MAPE (1.1791%)**.

Crop Yield Predictor | Course Project | Dr. M. Jenamani

[Jan'24 - Apr'24]

- Conducted data preprocessing and **EDA** on a dataset with **5000** rows, integrating weather, rainfall, and pesticides for model training.
- Engineered a crop yield prediction model, using **XGBoost** and **Random Forest** algorithms, and achieved an accuracy of **91.20%**.
- Created a user-friendly **GUI** using **Flask** to enable farmers to input data and receive real-time predictions, enhancing decision-making.

Brain Tumor Detection | Self Project

[Nov'23 - Dec'23]

- Detected the presence of a **brain tumor** and classified them into four classes, which includes no tumor, glioma, meningioma, and pituitary.
- Evaluated the performance of various **CNN** architectures, utilizing **PyTorch** and hyperparameter tuning to enhance model accuracy.
- Achieved an **F1 score** of **88.2%**, precision of 87.8% and **recall** of **88.4%**, ensuring accuracy in tumor detection and reducing false alarms.

Evacuation Models in Disaster Management | Course Project | Dr. M. Jenamani

[Aug'23 - Dec'23]

- Proposed an integrated **evacuation** and **distribution logistic system** to obtain simultaneous routing and scheduling of vehicles.
- Developed a **Mixed Integer Linear Programming** model to **minimize** vehicle arrival times at affected shelters and distribution centers.
- Implemented a **Memetic Algorithm** to optimize the efficient resolution of complex integrated evacuation and distribution issues.

TRAINING

Mahindra & Mahindra, Nagpur, Maharashtra

[Jun'22 - Aug'22]

- Obtained certification in the Vocational Training Program on Tractor Engineering conducted at Mahindra Excellence Centre.
- Acquired valuable insights into **supply chain operations** through shop floor visits, effectively fostering industry comprehension.
- Explored tractor components, understanding their industry contributions during shop floor visits, highlighting logistics integration.

SKILLS AND EXPERTISE

Programming Languages: Python | SQL | HTML | CSS | R **Tools:** Google Cloud Platform (BigQuery) | Power BI | Mongo DB | MS-Excel
Libraries: Dash | NumPy | Pandas | Matplotlib | Scikit-Learn | Seaborn | Plotly | Flask | Tensorflow | PyTorch | Keras

COURSEWORK INFORMATION

Applied Multivariate Statistical Modelling | Decision Modelling | Production Planning and Inventory Control | Safety & Risk Analytics | Statistical Decision Modelling | Supply Chain Management | Systems Modelling & Data Analytics | TQM and Six Sigma

POSITIONS OF RESPONSIBILITY

Secretary | Social & Cultural | SAM Hall of Residence | IIT Kharagpur

[Oct'23 - Apr'24]

- Coordinated a month-long preparation phase for the successful execution of "**Illumination 2023**", managing its planning and operation.
- Served as a crew member in Short Film Making, representing SAM Hall in the **General Championship** at IIT Kharagpur for the year 2024.
- Represented our hall in various General Championship events, including Debate and Elocution, alongside participants for the year 2024.

Teaching Assistant | Industrial & Systems Engineering | IIT Kharagpur

[Aug'24 - Present]

- Assisted in constructing **control charts** and using statistical tests like **ANOVA** to analyze process stability and measurement variability.
- Guided students in applying AI/ML architectural tools for quality analytics, including **object detection** and process capability analysis.

EXTRA CURRICULAR ACTIVITIES

- Represented Sir Ashutosh Mukherjee Hall of Residence in **Open IIT Vocals 2023**, earning a Certificate of Participation at IIT Kharagpur.
- Contributed significantly to **Optima 2023** as an Evaluation Committee member for the event "360 Degrees" during the departmental fest.
- Participated as a member of the **Inter-Hall Badminton** Competition for Sir Ashutosh Mukherjee Hall of Residence in the year 2024.